Alara Martin

alaram@berkeley.edu | linkedin.com/in/alara-martin/ | github.com/alaramartin | alaramartin.com

Education

University of California, Berkeley, B.A. Applied Math — Aug 2026 – May 2029

- **Relevant Coursework:** Abstract Linear Algebra, Multivariable Calculus
- **Clubs:** Education Team at Blockchain @ Berkeley, Treasurer at Cal Hiking and Outdoors Society

Foothill College, Dean's List (3.96 GPA) — Sep 2024 – Jun 2026

- **Relevant Coursework:** Advanced Data Structures & Algorithms in Java, Object-Oriented Programming in C++, Data Science, Discrete Math, Intro to Cybersecurity, Multivariable Differential Calculus

Palo Alto High School, 4.0 Unweighted / 4.48 Weighted GPA — Aug 2022 – Jun 2026

- **Relevant Coursework:** Linear Algebra, Real Analysis (at Stanford ULO), AP Physics C (Mech + E&M)
- **Awards:** AP Scholar with Distinction (3x), NSDA Speech & Debate Honor Society (<1%), President's Volunteer Gold

Skills

Languages & Libraries: Python, Java, C++, JavaScript, TypeScript, Solidity, HTML/CSS, LaTeX, PyTorch, React, Next.js
Tools & Concepts: Git, REST APIs, AWS, Firebase, Vercel, Tailwind CSS, Claude Code, UI Design

Experience

AI Research Intern — Jan 2025 – Present
Stanford University AI Development and Evaluation Lab — *Stanford, CA*

- Achieved 94% accuracy for medical imaging QA by building and training a Faster R-CNN keypoint detection model with a ResNet-50 backbone in PyTorch, reaching a median error of 1.7mm per point.
- Engineered a Python pipeline anonymizing 1,700+ DICOM images by parsing metadata tags to scrub patient fields.
- Co-authored paper (under review) releasing the dataset and model for developmental hip imaging QA.

Project Review Lead — Sep 2025 – Mar 2026
Hack Club Athena Initiative — *Palo Alto, CA*

- Reviewed 100+ technical projects and mentored participants in Git; built a tutorial website used by 1,500+.
- Ranked 24/1,500+ on the Athena coding leaderboard and selected as one of 200 hackathon invitees.

Club President & Outreach Founder — Sep 2023 – Jun 2026
Girls Who Code — *Palo Alto, CA*

- Developed project-based curriculum in web development and Python for 10+ peers; led a club robotics project.
- Founded a city-funded initiative teaching 15 elementary schoolers to code; 100% completed final projects.

Mathematics Tutor — Jan 2024 – Aug 2026
DreamCatchers (now Pathwise) & Operation Math — *Palo Alto, CA*

- Taught personalized algebra-to-geometry curriculum from diagnostic skill-gap assessments; improved scores 40%+.

Projects

4x Hackathon Winner — 2026

- **ORIGIN** (**1st Place Grand Prize, $4,000**), MantisGrid AI Infrastructure Hackathon: root-cause analysis agent for GPU cluster failures that builds a dependency graph from raw telemetry; beat the sponsor's solution at $0 inference cost.
- **SafePath** (**2nd Place Overall, $2,000**), Live Models Hackathon (Visko x Reactor): Next.js/FastAPI tool rendering any walking route as night-time video, fusing OpenStreetMap routing, Mapillary imagery, streetlight outages, and weather.
- **Cash Cow** (**2nd Most Creative, $500**), B@BHacks 2.0: accountability platform where users stake AUSD on their own deadlines and others bet on the outcome; AI agents verify completion directly from GitHub and Canvas.
- **UX Agent** (**1st, Best Use of TensorLake** & top 2 InsForge; top 8 of 70+), Nozomio Hackathon: AI agent pipeline that watches real PostHog feature analytics and autonomously ships UI updates informed by real user data.

PKL Viewer: Python, TypeScript, VS Code API - github.com/alaramartin/pkl-viewer — Aug 2025

- Developed a 31,100-user, open-source VS Code extension in Python and TypeScript enabling in-editor inspection of Python pickle files, adapting based on user feedback and collaborations through GitHub Issues/PRs.
- Designed a security-first default disassembly mode using pickletools (near-instantaneous) that resists malicious payloads by never executing code, plus an optional load for human-readable output.

DICOM Viewer & Editor: TypeScript, VS Code API - github.com/alaramartin/dicom-viewer — Aug 2025

- Built a VS Code extension with over 5,200 users across the VS Code Marketplace and Open VSX, extracting and rendering DICOM (.dcm) metadata tag trees directly in the editor view.
- Used native JS DICOM libraries for in-place metadata editing, enabling zero-setup installation with no external tools.